

Bruno Lauer

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EDUCATION

B.Eng. Electrical Engineering & Biomedical Engineering | *Western University* | London, ON | 2022 to 2026

- Cumulative GPA: ~3.8 | Dean's Honor List
- Relevant coursework: Power Systems, Electronics, Circuit Analysis, Embedded Systems, Control Systems, Biomedical Instrumentation, Biophysics, Cell Biology, Biochemistry, Molecular Biology, Schematic Design
- Laboratory training: cell culture and aseptic technique, PCR/qPCR, flow cytometry, ELISA, biological assays, quantitative data analysis and scientific reporting

Welding, Intermediate Certification | *Mohawk College* | Hamilton, ON | Jun 2022 to Aug 2023

- Completed welding training through intermediate level; practical fabrication and metalworking applicable to device prototyping, panel fabrication, and structural assembly

CERTIFICATIONS

AI Fluency: Framework and Foundations | Claude Academy, Anthropic | 2026

Structured training in applied AI concepts, prompt engineering, and responsible AI integration; applicable to ML-adjacent engineering roles and AI-assisted design and analysis workflows.

TECHNICAL SKILLS

Languages & Scripting: C/C++ (firmware, control loops, driver-adjacent systems), Python (ML pipelines, data analysis, automation), Java; Bash/shell scripting

Hardware & Electronics: PCB design (Autodesk EAGLE), analog front-end design, power electronics (MOSFETs, gate drivers, bootstrap circuits), Hall-effect sensing, op-amp signal conditioning, oscilloscope diagnostics, soldering

Embedded & Control: Arduino/AVR firmware, closed-loop current and position control, PWM generation, ADC sampling, servo and ESC integration

Graphics & Systems: Real-time 3D rendering pipeline architecture, anti-aliasing techniques (MSAA, TAA, DLSS/FSR), rasterization vs. ray-tracing, shader pipeline stages, FreeSync/HDR display standards, GPU driver architecture; low-level graphics and rendering pipeline exploration (independent study)

OS & Platform: Windows 10/11 (advanced configuration, driver management, registry, performance tuning), Linux (Ubuntu/Arch: CLI, kernel parameters, driver troubleshooting, package management)

Design & CAD: SolidWorks (3D modeling, assemblies, manufacturing drawings), AutoCAD (as-built documentation, structured DWG layers), 3D modeling (multiple packages)

Data & ML: scikit-learn, pandas, NumPy, matplotlib; supervised regression, feature engineering, cross-validation, model evaluation; AI-assisted workflows

Biomedical & Lab: Biomedical instrumentation, cell culture, aseptic technique, PCR/qPCR, flow cytometry, ELISA, cell counting, lab notebook documentation, data tracking and reporting

Productivity & Tools: MS Office Suite (Word, Excel, PowerPoint, Outlook), Git, Autodesk EAGLE, SolidWorks, AutoCAD

ENGINEERING PROJECTS

High-Current Buck Converter: Closed-Loop Power Supply (100A @ 1V)

Skills: PCB Design (EAGLE), Analog Circuit Design, Power Electronics, C Firmware, Closed-Loop Control, Oscilloscope Debugging, Signal Conditioning, Root-Cause Analysis, Technical Documentation

- Designed and built a complete 100A power electronics system from schematic through PCB layout, assembly, and performance validation: closed-loop current control in C on Arduino, Hall-effect sensing, op-amp signal conditioning, anti-aliasing RC filters, MOSFET switching with half-bridge gate driver
- Identified non-deterministic hardware PWM behavior via oscilloscope analysis, traced root cause, replaced with manual timer-controlled PWM, re-validated, and updated documentation throughout
- Achieved ~5% current measurement accuracy through calibration and moving-average filtering; debugged switching ripple through systematic inductance and layout analysis

BLDC Motor: Custom Design, Fabrication & Validation

Skills: SolidWorks, Electromechanical Design, Motor Control, Fabrication, Structured Testing, Iterative Design

- Designed a custom in-runner brushless DC motor in SolidWorks: selected pole/slot configuration to minimize cogging torque, modeled rotor/stator geometry, and generated manufacturing drawings
- Fabricated and assembled motor components with attention to magnetic alignment and mechanical balance; validated performance using an ESC module with structured torque/speed testing and iterative refinement

Solar Tracker: Embedded 2-Axis Gimbal System

Skills: Embedded Systems (Arduino), C Programming, Servo Control, Sensor Calibration, PID Tuning, Closed-Loop Control

- Designed and coded a 2-axis solar tracking system using servo motors and photoresistor feedback, achieving 82% tracking efficiency through iterative sensor calibration and PID parameter tuning
- Maintained structured test logs across all development cycles to guide systematic improvement

House Price Prediction: Supervised ML Pipeline (Python)

Skills: Python (pandas, NumPy, scikit-learn, matplotlib), Supervised Learning, Feature Engineering, Model Evaluation, Quantitative Reporting

- Built an end-to-end regression pipeline: data cleaning, feature engineering, model training and comparison (linear regression, random forest, gradient boosting), cross-validation, RMSE evaluation, and feature importance visualization

Assistive Inhaler Device: Multidisciplinary Biomedical Team Project

Skills: Human-Centred Design, Cross-Functional Collaboration, Systems Integration, Prototyping, Technical Presentation

- Co-led a full engineering design cycle for a patient-facing assistive device for visually impaired users: requirements definition, iterative prototyping, electrical/mechanical/software integration, and formal evaluation under competition constraints

WORK EXPERIENCE

Independent CAD & Technical Documentation Consultant | *Grand River Enterprises* | Ohsweken, ON | Jul 2026 to Sep 2026

- Engaged on a fixed-price contract to produce professional as-built CAD documentation for a multi-floor commercial facility; independently managed the full project lifecycle from site walkthrough and field measurements through to final client deliverables
- Produced scaled AutoCAD (DWG) floor plans with structured CAD layers incorporating room layouts, door/corridor placement, room numbering, and major architectural features suitable for facilities management and IT infrastructure planning; delivered finalized DWG and PDF packages alongside updated fire evacuation plans
- Managed client feedback across multiple revision cycles, maintained scope boundaries, and communicated technical deliverables directly to client throughout the engagement

General Laborer | *Abcott Construction* | Brantford, ON | May 2025 to Aug 2025

- Executed structural and civil construction tasks including foundation preparation, concrete forming/pouring, wood framing, and structural steel assembly; interpreted construction drawings daily and applied tolerance checks to verify dimensional accuracy and structural compliance
- Identified and resolved on-site build discrepancies independently before they became rework; coordinated work sequencing across multiple trades to maintain workflow efficiency, schedule, and site safety
- Contributed to material estimation, staging, and logistical planning; operated power tools, construction equipment, and heavy machinery under strict safety protocols

Low-Voltage Electrical Technician | *Cable Assembly LTD.* | Brantford, ON | Aug 2017 to Aug 2023

- Accumulated 2,000+ hours of hands-on electrical work across commercial construction sites: installed, terminated, and tested CAT6, CAT6A, and fiber optic networks to specification with rigorous quality inspection and documentation at every stage
- Led multiple projects end-to-end from scoping through final client sign-off; collaborated across multi-trade teams to ensure network compatibility, maintain cost-effectiveness, and uphold quality standards throughout

EXTRACURRICULAR & ADDITIONAL

- Independent graphics and rendering study: exploring low-level rendering pipeline architecture and geometry-shading interaction at the pipeline level; particular focus on real-time anti-aliasing, upscaling algorithms (DLSS/FSR), and GPU driver architecture
- Published free 3D assets to the design and engineering community via Gumroad; active contributor to online technical communities

- Independent fabrication and mechanical repair: carpentry, welding, heavy machinery, and automotive work; regularly assists community members with technical problem-solving
- Intramural Basketball | Short Filmmaking